

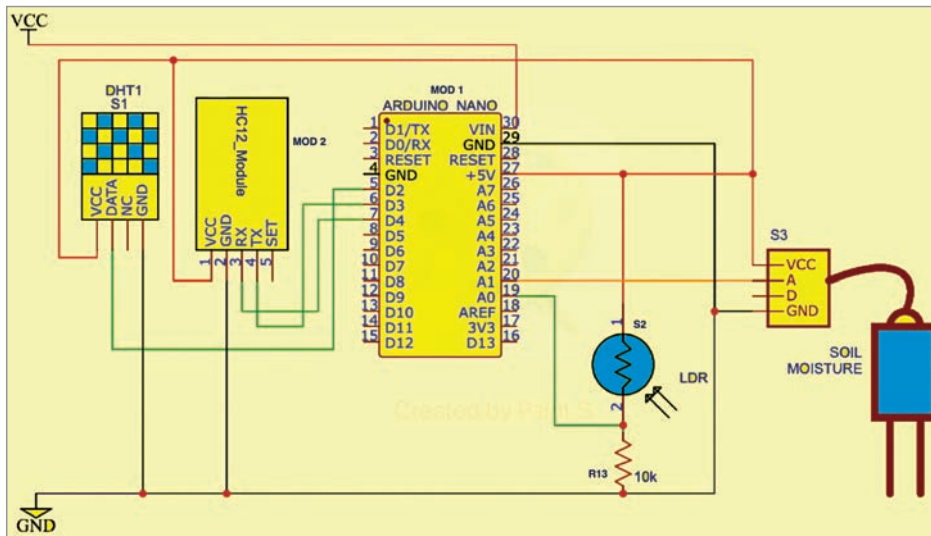


Fig. 4: Circuit diagram of the RDMS transmitter section

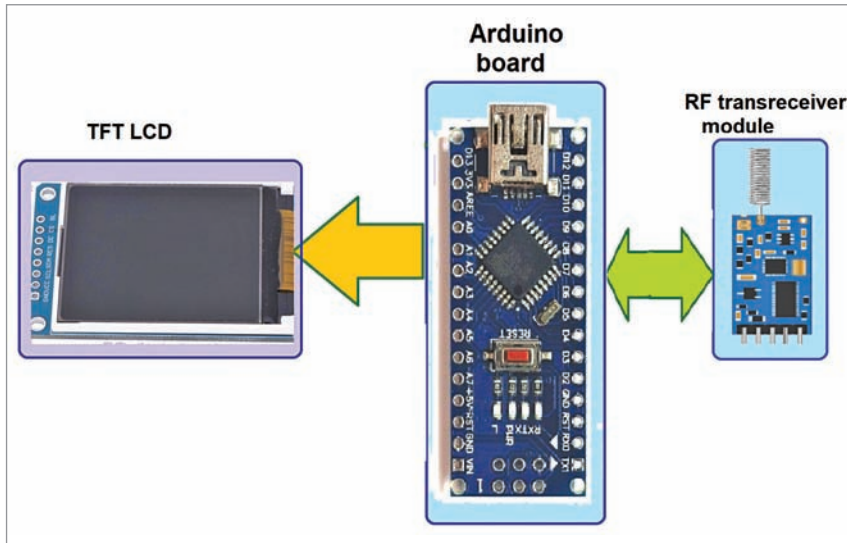


Fig. 5: Block diagram of the remote data receiver section

LCD monitor presents the temperature in degrees Celsius and humidity as a percentage of relative humidity (RH). Ambient light intensity and soil moisture level are both displayed as percentages on a 0-100 scale. The components utilised in this RDMS are detailed in Table 1.

The RDMS comprises both a data transmitter and receiver, as shown in Fig. 2. The transmitter includes one or more sensors (either identical or different), a microcontroller (MCU), and a radio transmitter. The

receiver features a radio receiver, an MCU, and a TFT LCD monitor.

Circuit and working

The operation of the remote data transmitter and receiver in the RDMS is described below.

RDMS transmitter. The remote data transmitter consists of three different sensors, an Arduino Nano board as the MCU, and an RF transceiver module. The primary components of the transmitter include the DHT11 sensor, LDR sensor, and

soil moisture sensor module, which measure temperature and humidity, ambient light, and soil moisture, respectively. The block diagram of the transmitter is shown in Fig. 3.

Circuit diagram of the RDMS transmitter. The circuit diagram of the data transmitter is presented in Fig. 4. The transmitter circuit is constructed using the DHT11, LDR, soil moisture sensor module, a 10-kilo-ohm resistor, an HC12 RF transceiver, and an Arduino Nano board (MOD1).

First, connect a 5V power supply to MOD1 externally after uploading the source code. Then, connect all components as shown in Fig. 4. The DHT11 sensor has three interfacing pins: VCC, GND, and DATA. The VCC pin connects to the 5V output pin of MOD1, the GND pin to the circuit ground, and the DATA output pin to the digital D2 pin of MOD1.

The soil moisture sensor also has three pins: VCC, GND, and A0. The VCC pin connects to the 5V output pin of MOD1, the GND pin to the circuit ground, and the A0 pin to the analogue input pin A1 of MOD1. The LDR is connected between the 5V power supply and analogue input pin A0, along with a 10-kilo-ohm pulldown resistor (R13).

The HC12 module has four interfacing pins: VCC, GND, Tx, and Rx. The VCC pin connects to the 5V output pin of MOD1, the GND pin to the circuit ground, and the Tx and Rx pins to the digital D3 and D4 pins of MOD1, respectively.

RDMS receiver. The receiver consists of an RF transceiver module, an Arduino Nano board as the MCU, and a TFT LCD monitor that displays all four parameters. Both the transmitter and receiver use Arduino Nano boards (MOD1,